



Homework 5

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**Optimal Classifier
Gaussian Classifier
Naïve Bayes Classifier**



Review of Bayes Theorem

Estimated time: 15min to watch the video + 10min for the questions

In the upcoming lecture, we will be looking in more depth at Classification Algorithms. Central to this is the Optimal Bayes Classifier, which uses Bayes Rules for classification purposes. You should already know Bayes rule from your math classes, but to have it fresh in your mind, let's review it.

- ♦ **Watch the video: “Bayes theorem, the geometry of changing beliefs”**

<https://www.youtube.com/watch?v=HZGCoVF3YvM>

This video introduces you to Bayes Rule.

- ♦ **Answer the following questions**

- 1) Describe in your own words the difference between $P(H)$ and $P(E)$ in Bayes Rule.
- 2) Why is the denominator $P(E)$ in Bayes Rule often replaced by a more complex formula and what is it replaced by?

1) given a sample set (100), $P(H)$ is the probability that a sample belongs to the H set, whereas $P(E)$ is the probability if the sample is carrying a specific characteristic E regardless of it it belongs to H or not

$$2) P(E) = P(H) \cdot P(E|H) + P(\text{not } H) \cdot P(E|\text{not } H)$$

$P(E)$ is the probability that a sample fits a certain evidence regardless of hypothesis in practice, it is often necessary to look at each set (H and not H) and within that set determine the probability that a sample fits E which is $P(H) \cdot P(E|H)$: probability that sample is H and fits E (is the same for not H)



Deeper Understanding of Bayes Theorem (OPTIONAL)

Estimated time: 21min to watch the video

- ♦ **Watch the video: “The medical test paradox, and redesigning Bayes' rule”**
<https://www.youtube.com/watch?v=IG4VkPoG3ko>

This video will lead you to a deeper understanding of Bayes rule. We do not absolutely need this for the upcoming class, but IMHO it is extremely helpful, so I highly recommend to watch it.



Recap – Linear Algebra

Estimated time: 70min to watch the recommended videos + whatever you decide to watch

Playlist “The Essence of linear Algebra”

https://www.youtube.com/watch?v=kjBOesZCoqc&list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab

This playlist consist of 11 basic chapters and a couple of footnotes, for a total 16 videos. Decide for yourself which videos you need to watch to recapitulate anything you may have forgotten.

I recommend this specific playlist as it is very graphical and non-formal, always putting the basic ideas first (that's why it's called the “Essence”), so it should be enjoyable to watch and easy to follow! I would recommend the watch the full playlist, but if you are firm in your math and want to skip some chapters, here are the most important ones:

- ♦ I would definitely recommend to watch chapters 1, 2, 3 and 4 if you are unsure about basis vectors and matrix multiplications. (*total time to watch: 40min*)
- ♦ You will need to know what the determinant (chapter 5) signifies and how to compute it. (*time to watch: 10min*)
- ♦ Furthermore, we will be using matrix inversion (chapter 6) and decomposition into eigenvectors (chapter 10). (*total time to watch: 30min*)



Review of Vector and Matrix Operation (linear Algebra)

Estimated time: 20min (more, if you have forgotten how to do this and need to lookup the calculations)

As we will be using the following in the upcoming lecture, make sure you know how to perform the following calculations.

Given 2-dimensional vectors $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ and 2x2-matrixes $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} e & 0 \\ 0 & f \end{bmatrix}$.

- a) Compute $u v^T$ and $u^T v$.
- b) Compute Au .
- c) Compute the determinantes $\det A$ and $\det B$.
- d) Compute the inverse matrix A^{-1} and B^{-1} .



Congratulations!

